

Cross-linguistic analysis of converbs: Semantic maps and parallel data for a better understanding of adverbial subordination in European languages

Initial dissimilarity analysis of syntactic forms

2026-06-07, 20:55

Spanish

Output files

This report generates a set of files. Each run produces its own files in a new subfolder. The subfolder is named with the target language and a random sequence of five characters (capital letters and digits). The names of all files within this subfolder are prefixed with the subfolder's name. For this run, the output file sequence is Spanish-LG2C3_ .

Research question

Can the participle in Spanish be used interchangeably with gerund? Model their usage on a parallel corpus (InterCorp):

- Record equivalent verb forms in the other languages.
- Use Multidimensional Scaling to create a two-dimensional space with the forms in the target language (Spanish), considering the syntactic forms of the equivalents in other languages. We adopt a Translation Mining script by Bert Le Bruyn (Le Bruyn et al. 2022).
- Examine the distribution of participles vs. gerunds in that space. When they separate into clusters, they are not interchangeable in Spanish.

Customized workflow

This workflow is customized to Spanish. When experimenting with other languages, use the French file as a default workflow. Changes: form.fr added,

Input

Ex post corrections of input data

```
# pozor, tady se sloupce nenahrazuji parametry, protoze jde o jazykove specificke upravu  
df$form.fr[df$form.fr == "adn"] <- "part" # dodatecna oprava vstupnich dat, mozna nebude pot
```

XLSX file with the following columns: ID_occurrence, src.lang, fr1, KWIC.fr, fr2, lemma.fr, meaning.fr, es, form.fr, form.cs, form.en, form.es, form.it, form.pl, form.ru.

Each row represents one sentence, where the French version uses either a participle or a gerund in the adverbial position.

The dataset is being continuously updated and may be incomplete at the time when this script is being executed. Therefore we always filter away incomplete observations; that is, we keep only rows that contain form labels for all investigated languages.

The current dataset has 364 observations (136 were filtered away as incomplete.)

Multidimensional Scaling

MDS with *form.<language>* as factors

Le Bruyn's function for [multidimensional scaling](#) ([Hamming distance](#) normalized to [0,1], two dimensions), slightly adapted and commented by us.

On input it requires a data frame containing just the “factors”, i.e. the *form.language* columns with the labels for the observed verb form in each language for the given item, and a vector of IDs. This vector actually corresponds to the ID_occurrence column in our input data frame.

```
custom_dissimilarity <- function(data, id_colname){  
  n <- nrow(data)  
  k <- ncol(data)  
  dist_matrix <- matrix(0,n,n) # an empty matrix (full of zeros)  
  for (i in 1:n) {  
    for (j in 1:n) {
```

```

    diff_count <- sum(data[i, ] != data[j, ])
    dist_matrix[i, j] <- diff_count / k } }
dist_df <- as.data.frame(dist_matrix)
rownames(dist_df) <- df[[id_colname]]
colnames(dist_df) <- df[[id_colname]]
return(dist_df)}

```

```

df_factors <- df %>%
  select(c(matches(regex("^form\\.(?!fr)")))))

dissimilarity_matrix <- custom_dissimilarity(data = df_factors,
                                             id_colname = params$row_id_column)
write_csv2(dissimilarity_matrix,
           file = str_c(outputfile_prefix,
                        "/",
                        outputfile_prefix,
                        "_distance_matrix.csv"),
           col_names = FALSE)

```

```

dist_matrix <- dissimilarity_matrix
dist_matrix <- dist_matrix
dist_object <- as.dist(dist_matrix)
mds_result <- cmdscale(dist_object, k = 2)
write_csv2(mds_result, file = str_c(outputfile_prefix,
                                     "/",
                                     outputfile_prefix, "_mds_result.csv"))

```

All analytical tables are saved in the folder Spanish-LG2C3_, along with the static plot below.

Semantic map (*form.<language>* as factors)

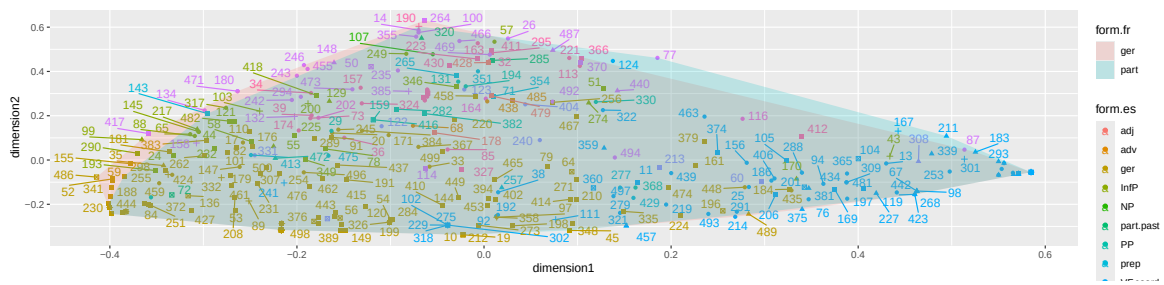
This plot shows each observation point

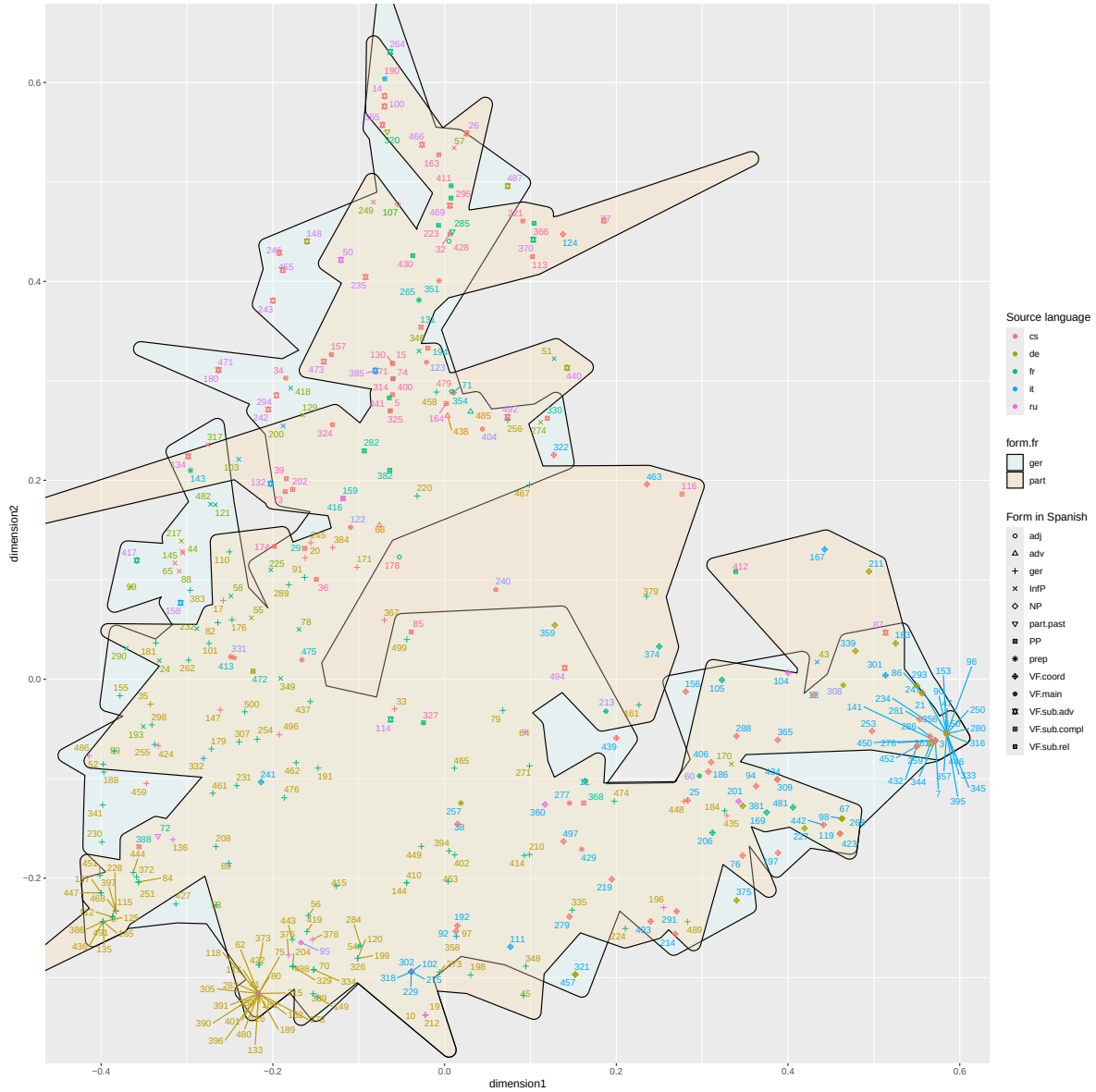
We use two color scales to contrast:

1. the Spanish form (Column form.es)
2. the source language of each item.

The numbers represent the IDs of the items and are slightly nudged from the points. When points overlap, their ID labels point to the same position.

Warning: ggrepel: 85 unlabeled data points (too many overlaps). Consider increasing max.overlaps





```
[[1]]
[1] "Spanish-LG2C3_/Spanish-LG2C3__semantic_map_plot.png"

[[2]]
[1] "Spanish-LG2C3_/Spanish-LG2C3__semantic_map_plot.pdf"

[[3]]
[1] "Spanish-LG2C3_/Spanish-LG2C3__semantic_map_plot.svg"
```



Figure 1: Linear models of contrast between the French and Spanish forms on two Spanish MDS dimensions. Whenever the Spanish forms map on both gerund and participle in French, those with participle in French tend to slightly higher values in both dimension. In all cases but VF.sub.rel, the difference is more pronounced in Dimension 2 than in Dimension 1. There was not enough data to show a trend on Spanish adj, adv, and NP. In this set of parallel sentences, the source language was not considered. French semantic labels are not included in the MDS dimensions.

Warning in qt((1 - level)/2, df): NaNs produced

Warning in qt((1 - level)/2, df): NaNs produced

Warning in max(ids, na.rm = TRUE): no non-missing arguments to max; returning -Inf

Warning in max(ids, na.rm = TRUE): no non-missing arguments to max; returning -Inf

Warning: ggrepel: 80 unlabeled data points (too many overlaps). Consider increasing max.overlaps

Warning: ggrepel: 21 unlabeled data points (too many overlaps). Consider increasing max.overlaps

Warning: ggrepel: 133 unlabeled data points (too many overlaps). Consider increasing max.overlaps

Warning: ggrepel: 14 unlabeled data points (too many overlaps). Consider increasing max.overlaps

Warning: ggrepel: 17 unlabeled data points (too many overlaps). Consider increasing max.overlaps

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[[1]]

[1] "Spanish-LG2C3_/Spanish-LG2C3__semantic_map_LM_plot.png"

[[2]]

[1] "Spanish-LG2C3_/Spanish-LG2C3__semantic_map_LM_plot.pdf"

[[3]]

[1] "Spanish-LG2C3_/Spanish-LG2C3__semantic_map_LM_plot.svg"

Interactive plots (*form.<language>* as factors)

These plots are only visible in documents rendered as html.

Documents in the target language, by their individual source languages

Hover your mouse over a point to see the text ID, semantic label, and concordance.

All documents together

Just documents with French as source language

MDS with *meaning.fr* and *form.<language>* as factors

```
df_factors_frmeaning <- df %>%
  #select(c(matches("^form\\.."), meaning.fr))
  select(c(matches(regex("^form\\.(!fr)")), meaning.fr))

dissimilarity_matrix_frmeaning <- custom_dissimilarity(data = df_factors_frmeaning,
write_csv2(dissimilarity_matrix_frmeaning,
  file = str_c(outputfile_prefix,
    "/",
    outputfile_prefix,
    "_distance_matrix_frmeaning.csv"),
  col_names = FALSE)
```

```
dist_matrix_frmeaning <- dissimilarity_matrix_frmeaning
#dist_matrix <- dist_matrix
dist_object_frmeaning <- as.dist(dist_matrix_frmeaning)
mds_result_frmeaning <- cmdscale(dist_object_frmeaning, k = 2)
write_csv2(mds_result_frmeaning, file = str_c(outputfile_prefix,
  "/",
  outputfile_prefix, "_mds_result_frmeaning.csv"))
```

All analytical tables are saved in the folder Spanish-LG2C3_, along with the static plot below.

Semantic map (*form.<language>* and *meaning.fr* as factors)

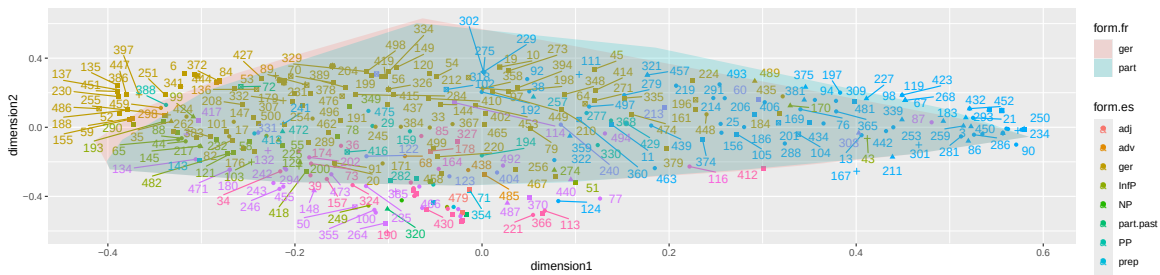
This plot shows each observation point

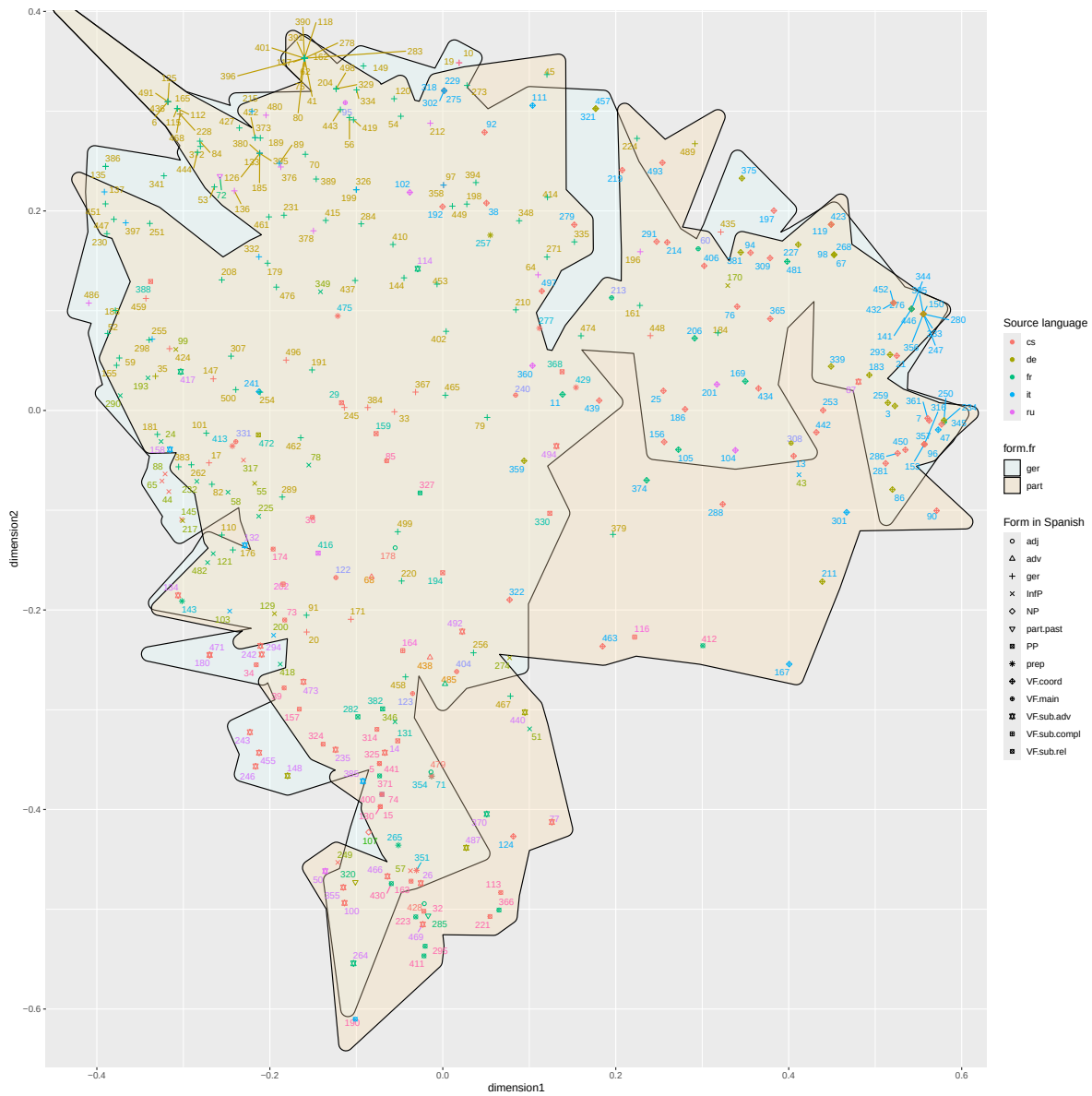
We use two color scales to contrast:

1. the Spanish form (Column *form.es*)
2. the source language of each item.

The numbers represent the IDs of the items and are slightly nudged from the points. When points overlap, their ID labels point to the same position.

Warning: ggrepel: 84 unlabeled data points (too many overlaps). Consider increasing *max.overlaps*





```
[[1]]
[1] "Spanish-LG2C3_/Spanish-LG2C3__semantic_map_frmeaning_plot.png"

[[2]]
[1] "Spanish-LG2C3_/Spanish-LG2C3__semantic_map_frmeaning_plot.pdf"

[[3]]
[1] "Spanish-LG2C3_/Spanish-LG2C3__semantic_map_frmeaning_plot.svg"
```

Interactive plots (*form.<language>* and *meaning.fr* as factors)

These plots are only visible in documents rendered as html.

Documents in the target language, by their individual source languages

Hover your mouse over a point to see the text ID, semantic label, and concordance.

All documents together

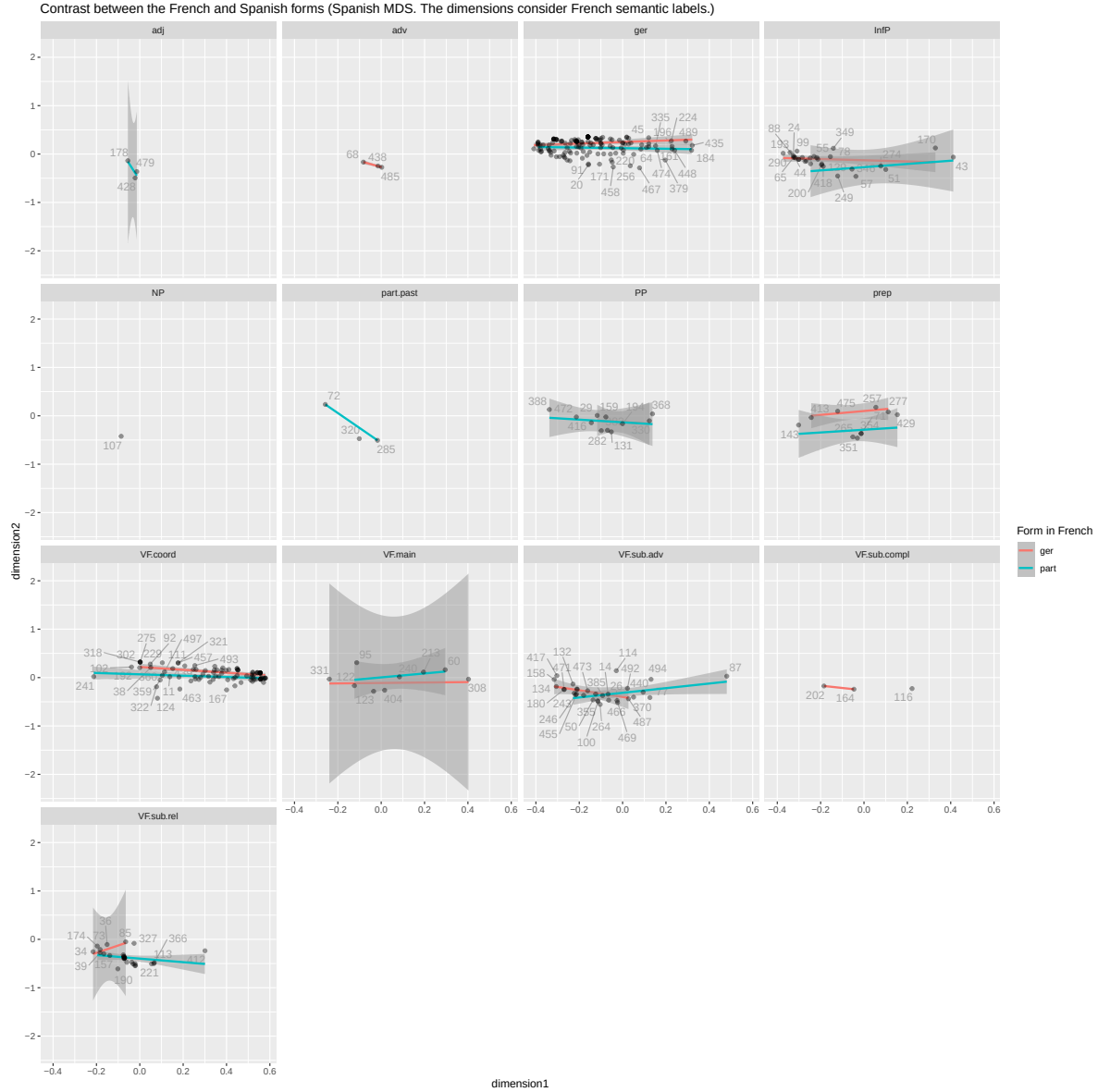


Figure 2: Linear models of contrast between the French and Spanish forms on two Spanish MDS dimensions. There was not enough data to show a trend on Spanish adj, adv, and NP. In this set of parallel sentences, the source language was not considered. French semantic labels are included in the MDS dimensions.

Warning in qt((1 - level)/2, df): NaNs produced

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Warning in max(ids, na.rm = TRUE): no non-missing arguments to max; returning

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Warning in max(ids, na.rm = TRUE): no non-missing arguments to max; returning
-Inf

Warning: ggrepel: 81 unlabeled data points (too many overlaps). Consider
increasing max.overlaps

Warning: ggrepel: 21 unlabeled data points (too many overlaps). Consider
increasing max.overlaps

Warning: ggrepel: 134 unlabeled data points (too many overlaps). Consider
increasing max.overlaps

Warning: ggrepel: 14 unlabeled data points (too many overlaps). Consider
increasing max.overlaps

Warning: ggrepel: 19 unlabeled data points (too many overlaps). Consider
increasing max.overlaps

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[[1]]

[1] "Spanish-LG2C3_/Spanish-LG2C3__LM_without_src_with_frmeaning.png"

[[2]]

[1] "Spanish-LG2C3_/Spanish-LG2C3__LM_without_src_with_frmeaning.pdf"

[[3]]

[1] "Spanish-LG2C3_/Spanish-LG2C3__LM_without_src_with_frmeaning.svg"

Just documents with French as source language

Linear regression model

A quickly drafted linear model (assumption unchecked, no model criticism): I try to predict dimension 2 from dimension 1 and the French semantic label. The model says: the higher dimension1, the lower dimension 2. At the same time, there is a statistically significant effect of attributive meaning (increasing the

```
glm(formula = dimension2 ~ dimension1 + meaning.fr,  
     data = data_frmeaning, family = "gaussian") %>%  
summary()
```

Call:

```
glm(formula = dimension2 ~ dimension1 + meaning.fr, family = "gaussian",  
     data = data_frmeaning)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.03062	0.05698	-0.537	0.5914
dimension1	-0.07283	0.03468	-2.100	0.0365 *
meaning.frattributive	-0.30071	0.06228	-4.829	2.06e-06 ***
meaning.frcause	-0.06430	0.06137	-1.048	0.2955
meaning.frconcess	-0.12476	0.07857	-1.588	0.1132
meaning.frconcom	0.23538	0.05888	3.998	7.81e-05 ***
meaning.frcond	-0.05831	0.17976	-0.324	0.7458
meaning.frdepictive	0.09080	0.10228	0.888	0.3753
meaning.frgram	-0.07842	0.13302	-0.590	0.5559
meaning.frmanner	0.07406	0.06194	1.196	0.2326
meaning.frposter	0.06482	0.11351	0.571	0.5684
meaning.frresult	-0.02255	0.08030	-0.281	0.7791
meaning.frsim	-0.04865	0.06224	-0.782	0.4350
meaning.frsimil/compar	-0.14382	0.17943	-0.802	0.4234

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.02892603)

Null deviance: 21.342 on 363 degrees of freedom
Residual deviance: 10.124 on 350 degrees of freedom
AIC: -240.95

Number of Fisher Scoring iterations: 2

Call:

```
glm(formula = dimension2 ~ meaning.fr, family = "gaussian", data = data_frmeaning)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.04254	0.05697	-0.747	0.456
meaning.frattributive	-0.28815	0.06229	-4.626	5.25e-06 ***
meaning.frcause	-0.04887	0.06123	-0.798	0.425
meaning.frconcess	-0.10761	0.07852	-1.370	0.171
meaning.frconcom	0.24263	0.05906	4.108	4.97e-05 ***
meaning.frcond	-0.03054	0.18014	-0.170	0.865
meaning.frdepictive	0.09921	0.10270	0.966	0.335
meaning.frgram	-0.08706	0.13360	-0.652	0.515
meaning.frmanner	0.09361	0.06153	1.521	0.129
meaning.frposter	0.05347	0.11393	0.469	0.639
meaning.frresult	-0.01301	0.08056	-0.161	0.872
meaning.frsim	-0.03459	0.06218	-0.556	0.578
meaning.frsimil/compar	-0.12842	0.18014	-0.713	0.476

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.02920694)

Null deviance: 21.342 on 363 degrees of freedom
Residual deviance: 10.252 on 351 degrees of freedom
AIC: -238.39

Number of Fisher Scoring iterations: 2

Check with a non-parametric method

```
kruskal.test(x = data_frmeaning$dimension2, g = data_frmeaning$meaning.fr)
```

Kruskal-Wallis rank sum test

data: data_frmeaning\$dimension2 and data_frmeaning\$meaning.fr
Kruskal-Wallis chi-squared = 187.81, df = 12, p-value < 2.2e-16

There are differences between groups. Between which groups can we observe significant differences and how do attributive meanings differ from other meanings? This table contains

only pairs with a statistical difference (Holm-adjusted p-value below 0.05) and pairs containing attributive meanings, sorted according to the adjusted p-value.

```
dunn <- dunn.test::dunn.test(x = data_frmeaning$dimension2, g = data_frmeaning$meaning.fr, n
tibble(dunn$comparisons, dunn$P.adjusted) %>%
  arrange(dunn$P.adjusted) %>%
  filter(str_detect(string = dunn$comparisons,
                    pattern = "attributive") | dunn$P.adjusted < 0.05 )
```

Effect sizes:

```
library(rcompanion)
epsilon2 <- rcompanion::epsilonSquared(x = data_frmeaning$dimension2, g = data_frmeaning$mean
epsilon2
```

	epsilon.squared	lower.ci	upper.ci
1	0.517	0.456	0.598

References

Le Bruyn, Bert, Martín Fuchs, Martijn van der Klis, Jianan Liu, Chou Mo, Jos Tellings, and Henriëtte de Swart. 2022. “Parallel Corpus Research and Target Language Representativeness: The Contrastive, Typological, and Translation Mining Traditions.” *Languages* 7 (3). <https://doi.org/10.3390/languages7030176>.